

Carbon isotopic composition, methanogenic pathway, and fraction of CH₄ oxidized in a rice field flooded year-round

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Received 22 February 2011; revised 17 August 2011; accepted 19 August 2011; published 8 December 2011.

[1] Values of $\delta^{13}\text{C}$ were investigated of CH₄ trapped in the soil pore water and floodwater of and emitted from a rice field under continuous flooding throughout the fallow and following rice seasons, and CH₄ produced via different pathways and fraction of CH₄ oxidized was calculated by using the isotopic data. Pore water CH₄ was relatively ^{13}C depleted, with $\delta^{13}\text{C}$ values about -65‰ over the season except between July and August (around -55‰). Also, hydrogenotrophic methanogenesis was very important (around 50%) for most of the season, while acetoclastic methanogenesis dominated (about 70%) only between July and August. Floodwater CH₄ was heavier in $\delta^{13}\text{C}$ value (from -50‰ to -34‰) than pore water CH₄ (from -68‰ to -54‰) over the season, demonstrating that it is highly influenced by methanotrophy. The $\delta^{13}\text{C}$ value of emitted CH₄ was negatively correlated with flux in temporal variation ($P < 0.05$), and it was more positive in the fallow season (between -56‰ and -44‰) than in the rice season (between -68‰ and -48‰). This indicates that plant-mediated CH₄ transport is probably a more important pathway and causes less CH₄ oxidation during the rice season than during the fallow season, which is further confirmed by the fraction of CH₄ oxidized being generally greater in the fallow season (60%–90%) than in the rice season (10%–80%). These findings suggest a low contribution of acetoclastic methanogenesis and a high fraction of CH₄ being oxidized in the field, especially in the fallow season.

Citation: Zhang, G., X. Zhang, Y. Ji, J. Ma, H. Xu, and Z. Cai (2011), Carbon isotopic composition, methanogenic pathway, and fraction of CH₄ oxidized in a rice field flooded year-round, *J. Geophys. Res.*, 116, G04025, doi:10.1029/2011JG001696.

1. Introduction

[2] The increase in concentrations of greenhouse gases (CO₂, CH₄, and N₂O) in the atmosphere is a main cause of global warming. CH₄ is the second largest radiative forcing of the long-lived greenhouse gas and 25 times greater than CO₂ in global warming potential on a 100 year horizon [Intergovernmental Panel on Climate Change (IPCC), 2007]. Over the last 40 years of the twentieth century, atmospheric CH₄ increased in concentration at an averaged rate of about 1% yr⁻¹. However, the rate has been decreasing substantially in the last 8 years [Dlugokencky *et al.*, 1998] and tended to be zero during 1999 to 2005 [Dlugokencky *et al.*, 2003]. The cause of neither the rapid increase nor the recent slowing down is clearly understood, which is directly related to the large uncertainties in the magnitudes, temporal and spatial distributions of identified CH₄ sources. As a consequence, considerable uncertainties remain in the global CH₄ budget. Rice fields are an important anthropogenic source of atmospheric CH₄, contributing about 31–112 Tg yr⁻¹ or 5%–19% to the total global emission [IPCC, 2007]. More information is hence needed about the source of CH₄ strength to

obtain a more accurate estimation of the global emission from the rice fields.

[3] Carbon isotope study is useful for quantifying CH₄ of different sources [Stevens and Engelkemeir, 1988; Miller, 2004]. Determining the carbon isotopic composition and fluxes of CH₄ is helpful in measuring relative source strengths of the CH₄ and providing constraints for the atmospheric CH₄ budget [Miller, 2004]. Studies of carbon isotopic composition of CH₄ from rice fields have been conducted, revealing that the values of $\delta^{13}\text{C}_{\text{CH}_4}$ varied greatly (Table 1). There are also two reports available in China. Li *et al.* [1997] reported that the $\delta^{13}\text{C}$ values of emitted CH₄ (just 6 gas samples) ranged from -65‰ to -54‰ . Moreover, Bergamaschi [1997] measured $\delta^{13}\text{C}$ values of CH₄ emitted directly from an intermittently irrigated rice field and CH₄ in gas bubbles collected from stirred paddy soil of the field. Both CH₄ were relatively depleted in ^{13}C , with $\delta^{13}\text{C}$ values ranging from -71‰ to -52‰ and -71‰ to -58‰ , separately. No comparable study, however, has so far been done throughout the fallow and following rice seasons, related to the continuously flooded rice fields, a special kind of rice fields in China. Because of various reasons, such as poor drainage conditions caused by side leaching invasion in hilly and mountainous areas or very high groundwater table in plain, and poor irrigation conditions which could not ensure enough water for rice transplanting next year, about 12% of the rice fields in China is flooded all year-round [Cai, 1999; Cai *et al.*, 2003].

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Table 1. Overview of Available Carbon Isotopic Composition of CH_4 From Rice Field^a

Location	CF ^b or Not	Sampling Method/Type	Observation Period	$\delta^{13}\text{C}_4$ (‰)	Reference
USA	NA ^b	Nocturnal inversion	3 weeks before harvest	-68 to -66	<i>Stevens and Engelkemeir</i> [1988]
Kenya	NA	Static chamber	Days 44, 72, 100, 128, 156 after harvest	-63 to -57	<i>Tyler et al.</i> [1988]
USA	NA	NA	May–June	-66 to -60	<i>Wahlen et al.</i> [1989]
Italy	NA	Static chamber	Days 85–88 after flooding	-68 to -63	<i>Bergamaschi</i> [1990]
Japan	NA	NA	NA	-60 to -52	<i>Nakamura et al.</i> [1990]
Japan	No	Gas bubbles ^c	Throughout the 1989 rice season	-68 to -48	<i>Uzaki et al.</i> [1991]
NA	NA	NA	NA	-65 to -61	<i>Wahlen</i> [1993]
Japan	Yes	Static chamber	Days 22, 51, 71, 93, 107 in 1990 and Days 38, 72, 100 in 1991 after flooding	-71 to -56 -72 to -58	<i>Tyler et al.</i> [1994]
China	No	Static chamber	Throughout the 1995 rice season	-71 to -52	<i>Bergamaschi</i> [1997]
		Gas bubbles ^c	Throughout the 1995 rice season	-71 to -58	
USA	Yes	Controlled chamber	Days 84–89 after seeding	-66 to -61	<i>Chanton et al.</i> [1997]
		Gas bubbles ^c	Days 84 after seeding	-61 to -57	
		Lacunal gas	Days 84 after seeding	-56 to -50	
		Pore water	Days 84 after seeding	-70 to -59	
		Floodwater	Days 84 after seeding	-46 to -42	
Japan	Yes	Static chamber	16 June to 15 Sep	-70 to -37	<i>Chidthaisong and Watanabe</i> [1997]
		Aerenchyma gas	Days 76, 83, 99, 105, 112 after flooding	-41 to -35	
China	NA	Static chamber	26 Aug, 3 Sep, 9 Sep	-65 to -54	<i>Li et al.</i> [1997]
USA	Yes	Static chamber	Days 23, 38, 56, 77, 86 after flooding	-58 to -53	<i>Tyler et al.</i> [1997]
		Gas bubbles ^d	Days 79, 86 after flooding	-49 to -43	
		Aerenchyma gas	Days 51, 79 after flooding	-43 to -41	
		Pore water	Days 37, 51, 63, 72, 86 after flooding	-61 to -48	
USA	Yes	Static chamber	Throughout the 1995 rice season	-63 to -46	<i>Bilek et al.</i> [1999]
		Pore water	Days 22, 36, 49, 65 after flooding	-57 to -48	
Italy	No	Static chamber	Days 71–81 after flooding	-69 to -64	<i>Dan et al.</i> [2001]
Italy	No	Static chamber	Days 33, 57, 75, 95, 112 after flooding	-68 to -60	<i>Krüger et al.</i> [2001]
Italy	No	Gas bubbles ^c	Days 33, 57, 75, 95, 112 after flooding	-61 to -54	<i>Krüger et al.</i> [2002]
		Gas bubbles ^d	Days 33, 57, 75, 95, 112 after flooding	-58 to -48	
		Aerenchyma gas	Days 31, 55, 73, 91, 112 after flooding	-53 to -49	
		Pore water	Days 33, 57, 75, 112 after flooding	-60 to -40	
Italy	No	Controlled chamber	Throughout the 1998 rice season	-67 to -47	<i>Marik et al.</i> [2002]
		Gas bubbles ^d	Throughout the 1999 rice season	-65 to -53	
Thailand	No	Gas bubbles ^c	Mineral soil: Dec 1990 to Mar 1992	-63 to -31	<i>Nakagawa et al.</i> [2002]
			Peat soil: Dec 1990 to Mar 1992	-61 to -45	
Italy	Yes	Static chamber	Throughout the 1999 rice season	-73 to -58	<i>Krüger and Frenzel</i> [2003]
		Aerenchyma gas	Days 105 after flooding	-50	
		Pore water	Days 0, 4, 24 after final fertilization	-66 to -53	
		Gas bubbles ^c	Days 5 after final fertilization	-59 to -56	
		Gas bubbles ^d	Days 5 after final fertilization	-57 to -55	
Japan	Yes	Pore water	Throughout the 2002 rice season	-70 to -40	<i>Cheng et al.</i> [2005]
		Floodwater	Throughout the 2002 rice season	-60 to -20	
Germany	Yes	Static chamber	Days 15–85 after transplanting	-68 to -61	<i>Conrad and Klose</i> [2005]
		Pore water	Days 15–85 after transplanting	-55 to -46	
Japan	No	Emitted from soil surface	Days 104–132 after flooding	-70 to -13	<i>Han et al.</i> [2005]
		Aerenchyma gas	Days 105, 113, 124 after flooding	-59 to -54	
		Entrapped in soil	Days 104–132 after flooding	-71 to -41	
India	No	Static chamber	Throughout the 2000 rice season	-60 to -46	<i>Rao et al.</i> [2008]
			Throughout the 2003 rice season	-59 to -46	
China	Yes	Static chamber	Throughout the 2008 fallow season	-56 to -44	<i>This study</i>
			Throughout the 2009 rice season	-68 to -48	
		Pore water	Throughout the 2008 fallow season	-68 to -60	
		Floodwater	Throughout the 2009 rice season	-68 to -54	
			Throughout the 2008 fallow season	-45 to -34	
			Throughout the 2009 rice season	-50 to -40	

^aPartially adopted from the study of *Marik et al.* [2002].^bCF, continuous flooding; NA, information not available.^cStirred up.^dNaturally released.

Continuous flooding leads soils to intensive anaerobicity and high CH_4 emission, and the contribution of such fields to the total CH_4 emission is over 40% [*Cai, 1999; Cai et al., 2003*]. It is, therefore, essential to investigate rules of the CH_4 emissions and corresponding ^{13}C -isotopic signatures in particular.

[4] Measurement of carbon isotopic ratios is also useful for identifying processes of CH_4 emission including production, oxidation and transport of CH_4 from rice fields. It is principally possible to use values of ^{13}C measured of CH_4 , CO_2 and acetate to calculate relative contribution of each pathway to the total CH_4 production and fraction of the CH_4

oxidized [Tyler *et al.*, 1997]. In rice paddies, CH_4 is formed mainly by way of acetate fermentation and CO_2 reduction [Takai, 1970]. However, the relative contribution of the two methanogenic pathways to total CH_4 production can vary considerably from one environment to the other, such as different temperatures [Fey *et al.*, 2004] or different habitats [Conrad, 1999; Conrad *et al.*, 2002] and soils [Tyler *et al.*, 1997; Yao and Conrad, 2000; Nakagawa *et al.*, 2002], and is significantly affected by rice planting and rice cultivar [Tyler *et al.*, 1997; Bilek *et al.*, 1999]. Conversely, methanogenic pathways and effects of CH_4 oxidation and transport are crucial for the extent of carbon isotope fractionation. Methanogenesis via CO_2 reduction exhibits a much stronger fractionation factor than that via acetate fermentation [Whiticar, 1999], and methanotrophic bacteria consume $^{12}\text{CH}_4$ faster than they do $^{13}\text{CH}_4$, thus leaving the residual CH_4 enriched in ^{13}C [Whiticar, 1999; Venkiteswaran and Schiff, 2005]. Furthermore, as $^{12}\text{CH}_4$ is more readily emitted than $^{13}\text{CH}_4$, the remaining CH_4 dissolved in the water becomes ^{13}C enriched, as well as CH_4 in the aerenchyma and the rhizosphere [Popp *et al.*, 1999; Chanton, 2005].

[5] CH_4 in soil pore water as a whole poses as a large CH_4 reservoir in the paddy soil, contributing to CH_4 emission from the rice fields [Alberto *et al.*, 2000]. The production zone of CH_4 in the bulk soil is considered to be the origin of CH_4 in the paddy soil to some extent [Tyler *et al.*, 1997; Bilek *et al.*, 1999]. CH_4 in floodwater, another kind of CH_4 entrapped in rice fields and transported through the oxidizing layer, namely, the soil-water interface, is supposed to be highly affected by methanotrophy [Tyler *et al.*, 1997]. Measuring $\delta^{13}\text{C}$ values of CH_4 in various field compartments simultaneously should help us better understand the processes of CH_4 emissions from rice fields. In comparison of rice-growing season, on the other hand, it is expected to find large differences in pathways of CH_4 production as well as CH_4 oxidation ratio in flooded rice fields all year-round, owing to no rice planting and relative low temperature in the fallow season.

[6] For these purposes, a field experiment was conducted from November 2008 to November 2009 in China to investigate carbon isotopic signatures of CH_4 in soil pore water and floodwater of and emitted from a rice field under continuous flooding throughout the fallow and following rice seasons, and then to report its methanogenic pathway and fraction of the CH_4 oxidized.

2. Methods and Measurements

2.1. Field Description and Experiment Design

[7] The field experiment was laid out with plots in triplicate, 3×4 m each, in a paddy field at Baitu Town, Jurong City, Jiangsu Province, China ($31^\circ 58'\text{N}$, $119^\circ 18'\text{E}$). The soil of the rice field is classified as Typic Haplaquepts (USDA, 1975). Soil samples were collected from the top layer (0–0.15 m) for measurement of initial organic C, total N and isotopic signature of the soil organic carbon, which was found to be 0.96%, 0.10% and -27.4% , respectively.

[8] After the rice in the plots was harvested on 2 November 2008, rice stubbles and weeds were all removed from the plots. Then the plots were kept flooded from 18 November 2008 to 15 October 2009 and drained prior to the next rice harvest on 16 October 2009. In 2009, all the plots were

ploughed the way the local farmers do before rice transplanting. Rice seeds (Cultivar “*Oryza sativa* L. Huajing 3”) were sown in nursery beds on 25 May, seedlings transplanted into the plots on 26 June at their 3 to 4 leaf stage, and the crop harvested on 3 November. Urea was applied at a rate of 300 kg N ha^{-1} , 50% as basal fertilizer on 26 June, 25% as tillering fertilizer on 17 July, and 25% as panicle fertilizer on 16 August. Both calcium superphosphate and potassium chloride were applied as basal fertilizer on 26 June at a rate of 450 kg ha^{-1} and 225 kg ha^{-1} , respectively.

2.2. Field Sampling and Measuring

[9] The static chamber method was used to sample CH_4 flux. Plastic bases for the chambers ($0.5 \times 0.5 \times 1$ m each) were installed on 12 November 2008 and kept there until rice harvest in November 2009. Removable wooden boardwalks (2 m long) were set up on 18 November 2008 to avoid disturbance of the soil during sampling for measurement. To measure CH_4 flux, four gas samples were collected using 18 mL vacuum vials from each chamber at 15 min intervals during the period between 08:00 A.M. and 12:00 noon on each sampling day. The collection was repeated once every 7–10 days in the fallow season and every 4–7 days during the rice season. CH_4 flux was figured out based on the linear increase in gas concentration of the samples with each sampling and adjusted in light of the volume and the bottom area of the sampling chamber. Mean CH_4 flux is the average of the triplicates weighted by the interval between two adjoining measurements. Soil temperature at 0.1 m depth was measured on each sampling day with a hand-carried digital thermometer (Yokogawa Meter and Instruments Corporation, Japan).

[10] After flux of CH_4 became detectable, two gas samples were taken at 10–15 day intervals with special sampling bags (Aluminum foil compound membrane, 0.5 L, Delin gas packing Co., Ltd, Dalian, China) using a small battery-driven pump for carbon isotopic composition measurement. After the final drainage, sampling at 1–2 day intervals was conducted to measure CH_4 flux and corresponding $\delta^{13}\text{C}$ value. The first sample was taken directly after closure of the chambers, while the second was taken at the end of the 1–2 h closure period. Isotopic signature (S) of the emitted CH_4 was calculated as follows:

$$S = [(B \times b) - (A \times a)] / (B - A), \quad (1)$$

where A and B stands for CH_4 concentrations ($\mu\text{L L}^{-1}$) in the samples at the beginning and at the end, respectively, of the sampling period and a and b for corresponding $\delta^{13}\text{C}_4$ values (‰) of the gas samples.

[11] Soil pore water samples were collected at 15–30 day intervals using a Rhizon soil moisture sampler (10 RHIZON SMSMOM, Eijkelkamp Agrisearch Equipment, Giesbeek, Netherlands), which consists of a 0.1 m long microporous polymer tube ($2.5 \text{ mm OD} \times 1.5 \text{ mm ID}$) and a 1 m long PVC hose ($2.7 \text{ mm OD} \times 1 \text{ mm ID}$). The microporous polymer tube was buried horizontally in the soil, 0.1 m in depth, in each plot on 18 November 2008 and then left there throughout the fallow and following rice seasons. Prior to sampling, about 5 mL soil solution was extracted using 18 mL vacuum vial to flush and purge the sampler. Then approximately 10 mL of soil solution was drawn into another

vial for analysis. Simultaneously, 10 mL of floodwater was collected using a plastic syringe and transferred into a 18 mL vacuum vial. Finally, the pressure in all the vials was equilibrated by filling in pure N_2 gas. After vigorous shaking by hand, the air in the headspace of the vials was directly analyzed for CH_4 on a GC-FID. Their corresponding $\delta^{13}\text{CH}_4$ values were then determined using an isotope ratio mass spectrometer. CH_4 concentration (C_{CH_4}) in pore water and floodwater was calculated using the following equation:

$$C_{\text{CH}_4} = (m \times G_V) / (G_L \times M_V) (\mu\text{M L}^{-1}) \quad (2)$$

where m stands for mixing ratio of CH_4 in the headspace of the vial ($\mu\text{L L}^{-1}$), M_V for gas volume of an ideal gas (24.78 L mol^{-1} at 25°C), G_V for volume of the headspace of the vial (L), and G_L for volume of the liquid in the vial (L).

2.3. Measurement of Gas Concentrations and Isotope Ratios

[12] Concentration of CH_4 was analyzed with a gas chromatograph (Shimadzu GC-12A, Kyoto, Japan) equipped with a flame ionization detector (FID) and a 2 m Poropak Q (80/100 mesh) column [Cai *et al.*, 2000]. The carbon isotopic composition was determined using the continuous flow technique on a Finnigan MAT 253 isotope ratio mass spectrometer (Thermo Finnigan, Bremen, Germany), which has a fully automated interface for pre-GC concentration (PreCon) of trace gases [Cao *et al.*, 2008]. CH_4 in gas samples, pre-concentrated on the PreCon by He (carrier gas, 99.999% in purity, at a rate of 20 mL min^{-1}) and then converted to CO_2 in a combustion furnace at 1000°C , was flowed into GC for separation. After CO_2 was separated from other components in GC, it went into the mass spectrometer for analysis of isotopic signature. CO_2 reference gas (99.999% in purity, with a $\delta^{13}\text{C}_{\text{PDB}}$ value of -23.73‰) was injected 3 times into the ionic source continuously, at 30 s intervals. The CO_2 peak occurred at about 870 s and the ratio line was a positive peak. On the basis of the ratios of the CO_2 peak of the working standard to the intensity of the three ionic flows, $\delta^{13}\text{C}$ of CO_2 derived from the CH_4 in the samples was figured out relative to the international carbon isotope standard PDB ($\delta^{13}\text{C}_{\text{PDB}}\text{‰}$). The compressed air with $2.02 \mu\text{L L}^{-1}$ CH_4 was regarded as the sample from a stable source, and the standard deviations of $\delta^{13}\text{C}$ was $\pm 0.196\text{‰}$ ($n = 9$). Isotope ratios were expressed in the standard delta notation:

$$\delta^{13}\text{C} = (R_{\text{sample}}/R_{\text{standard}} - 1) \times 1000(\text{‰}) \quad (3)$$

where R_{sample} and R_{standard} denotes the ratio of $^{13}\text{C}/^{12}\text{C}$ in the sample and in standard methane, respectively.

2.4. Calculations

[13] The contribution of acetate to the total CH_4 production (f_{ac}) is calculated by the following mass balance, assuming that either the acetate or H_2/CO_2 is the substrate in rice fields [Tyler *et al.*, 1997; Bilek *et al.*, 1999; Krüger *et al.*, 2002]:

$$f_{\text{ac}} = \text{CH}_4(\text{acetate}) / (\text{CH}_4(\text{acetate}) + \text{CH}_4(\text{H}_2/\text{CO}_2)) \quad (4)$$

$$\delta^{13}\text{CH}_4 = f_{\text{ac}} \times \delta^{13}\text{CH}_4(\text{acetate}) + (1 - f_{\text{ac}}) \delta^{13}\text{CH}_4(\text{H}_2/\text{CO}_2) \quad (5)$$

where $\delta^{13}\text{CH}_4(\text{acetate})$ and $\delta^{13}\text{CH}_4(\text{H}_2/\text{CO}_2)$ stands for $\delta^{13}\text{C}$ values of CH_4 produced by acetate and H_2/CO_2 , respectively, and $\delta^{13}\text{CH}_4$ for $\delta^{13}\text{C}$ value of the total CH_4 production.

[14] Carbon isotope fractionation factor for acetate or H_2/CO_2 producing CH_4 is defined according to Hayes [1993] by:

$$\epsilon_{\text{acetate}/\text{CH}_4} = (1 - \alpha_{\text{acetate}/\text{CH}_4}) 1000 \approx \delta^{13}\text{CH}_4(\text{acetate}) - \delta^{13}\text{C}_{\text{acetate}} \quad (6)$$

$$\alpha_{\text{CO}_2/\text{CH}_4} = (\delta^{13}\text{CO}_2 + 1000) / (\delta^{13}\text{CH}_4(\text{H}_2/\text{CO}_2) + 1000) \quad (7)$$

where $\delta^{13}\text{C}_{\text{acetate}}$ stands for $\delta^{13}\text{C}$ value of acetate, $\delta^{13}\text{CH}_4(\text{acetate})$ value is in the range from -36‰ to -48‰ [Conrad *et al.*, 2002], and $\alpha_{\text{CO}_2/\text{CH}_4}$ is found to be $1.025\text{--}1.083$ [Fey *et al.*, 2004].

[15] Additionally, the fraction of CH_4 that is oxidized (f_{ox}) in rice fields is estimated as follows [Stevens and Engelkemeir, 1988; Tyler *et al.*, 1997]:

$$f_{\text{ox}} = (\delta^{13}\text{CH}_4(\text{original}) - \delta^{13}\text{CH}_4(\text{oxidized})) / [(1/\alpha_{\text{ox}} - 1) \times (\delta^{13}\text{CH}_4(\text{oxidized}) + 1000)] \quad (8)$$

where $\delta^{13}\text{CH}_4(\text{original})$ stands for carbon isotopic composition of primarily produced CH_4 , $\delta^{13}\text{CH}_4(\text{oxidized})$ for carbon isotopic composition of residual CH_4 after methanotrophy, which is calculated in terms of a semiempirical equation [Tyler *et al.*, 1997]:

$$\delta^{13}\text{CH}_4(\text{oxidized}) = \delta^{13}\text{CH}_4(\text{emitted}) - \epsilon_{\text{transport}} \quad (9)$$

and α_{ox} for isotope fractionation factor, which is consumed by methane-oxidizing bacteria. Generally, α_{ox} is $1.013\text{--}1.049$ [Zhang *et al.*, 2009].

2.5. Statistical Analysis

[16] Statistical analysis was conducted with the aid of Software SPSS 15.0 for Windows (SPSS Inc., Chicago), and Standard deviation of means was calculated with software Microsoft Excel 2003 for Windows. Sample sets were rejected unless they yielded a linear regression value of $r^2 \geq 0.90$.

3. Results

3.1. CH_4 Concentration and Corresponding $\delta^{13}\text{CH}_4$ and $\delta^{13}\text{CO}_2$ in Pore Water

[17] CH_4 concentration in pore water increased obviously in the fallow season and peaked on 5 June (Figure 1a). After rice transplanting, it descended markedly but soon turned upward to a second peak on 12 October. It ranged from 6 to $398 \mu\text{M L}^{-1}$ in the fallow season and from 21 to $336 \mu\text{M L}^{-1}$ during the following rice season. Pore water CH_4 was relatively constant in $\delta^{13}\text{C}$ value, around -65‰ in the early fallow season but increased to -60‰ in the late period (Figure 1b). After rice transplanting, the value increased markedly again to about -55‰ in the beginning and nearly unchanged between July and August. Nevertheless, it declined rapidly to -68‰ at the end. In general, pore water CH_4 was relatively ^{13}C depleted, with $\delta^{13}\text{C}$ values about -65‰ for the most of the season. CO_2 in pore water exhibited a rather heterogeneous temporal variation pattern in $\delta^{13}\text{C}$ value

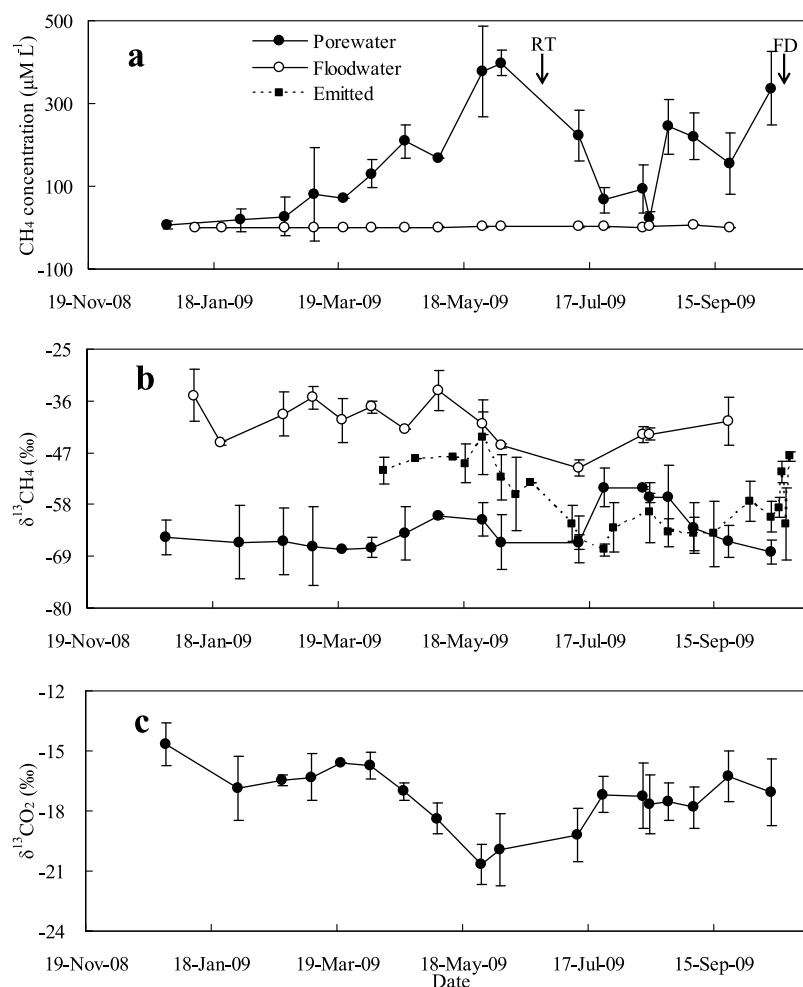


Figure 1. Temporal variations of (a) CH_4 concentration and (b) corresponding $\delta^{13}\text{CH}_4$ in soil pore water and floodwater and $\delta^{13}\text{C}$ -values of emitted CH_4 , and (c) of the $\delta^{13}\text{CO}_2$ in soil pore water from November 2008 to November 2009. RT, rice transplanting; FD, final drainage.

(Figure 1c). In the fallow season, it gradually became ^{13}C depleted and its $\delta^{13}\text{C}$ value decreased from -15‰ to -21‰ . However, in the rice season, the value increased again to -16‰ .

3.2. CH_4 Concentration and Corresponding $\delta^{13}\text{CH}_4$ in Floodwater

[18] As expected, CH_4 concentration in floodwater was very low ($<7 \mu\text{M L}^{-1}$) over the season and was significantly lower than that in pore water (Figure 1a). CH_4 in floodwater was also measured for $\delta^{13}\text{C}$ value (Figure 1b). After a short ^{13}C depletion in the fallow season, floodwater CH_4 first became heavier, with $\delta^{13}\text{C}$ values from -45‰ to -34‰ . Subsequently, the value decreased to -45‰ at the end. After rice transplanting, it again increased during the season from -50‰ to -40‰ . Obviously, floodwater CH_4 was more ^{13}C enriched in contrast to pore water CH_4 during the whole observational period (Figure 1b).

3.3. CH_4 Emission and Corresponding $\delta^{13}\text{CH}_4$

[19] Pronounced temporal variation of CH_4 flux was observed (Figure 2). In the fallow season, a large CH_4 flux was observed just 20 days before rice transplanting.

Afterward, it increased quickly at the beginning of the rice season and varied significantly, peaking ($23.1\text{--}29.8 \text{ mg CH}_4 \text{ m}^{-2} \text{ h}^{-1}$) twice in July and August, separately. Final drainage before rice harvest led to a brief increase in flux, which was followed by a rapid decrease, probably because of the fact that the CH_4 stored in the soil got released after disappearance of the floodwater layer (Figure 1a). The total CH_4 emission in the fallow season was $3.3 \text{ g CH}_4 \text{ m}^{-2}$, accounting for 8.9% of the annual total emission. Significant changes in soil temperature were also recorded (Figure 2), with a mean value of 13.2°C in the fallow season and 24.4°C during the rice season. The flux tallied well with soil temperature in temporal variation, and statistical analysis also showed a significant positive correlation between the two over the season ($r = 0.635$ and 0.732 , respectively, $P < 0.01$).

[20] Similar to the flux of CH_4 , $\delta^{13}\text{C}$ value of emitted CH_4 also varied apparently as is shown in Figure 1b. Since little flux was detected until April 2009, measurement of $\delta^{13}\text{CH}_4$ began in the late fallow season. Before rice transplanting, it increased from -51‰ to -44‰ at the beginning and then dropped rapidly at the end. After rice transplanting, the emitted CH_4 was ^{13}C depleted with $\delta^{13}\text{C}$ values being around -65‰ in most of the rice season. Late in the rice

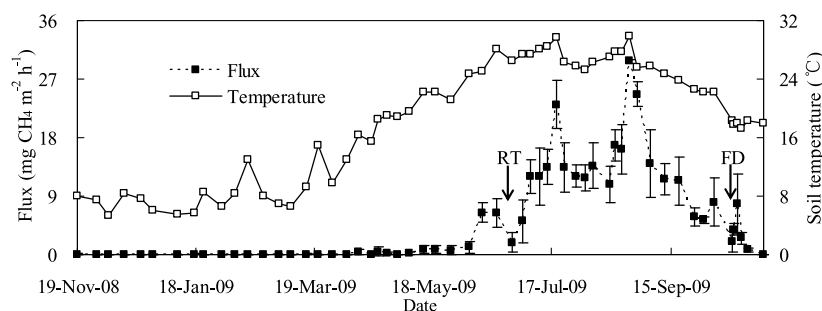


Figure 2. Temporal variations of (a) CH_4 flux and (b) soil temperature from November 2008 to November 2009. RT, rice transplanting; FD, final drainage.

season, however, the emitted CH_4 became heavier. The impact of the final drainage on $\delta^{13}\text{C}$ value of emitted CH_4 was remarkable (Figure 1b). Emitted CH_4 was significantly enriched in ^{13}C 1 day after the final drainage, with $\delta^{13}\text{C}$ value increased markedly to -51‰ . The value declined rapidly to -62‰ just 2 days later but ascended again to -48‰ in the 2 days that followed. Taken as a whole, the $\delta^{13}\text{C}$ value of emitted CH_4 was more positive in the fallow season (from -56‰ to -44‰) than in the following rice season (from -68‰ to -48‰). Notably, temporal variation of CH_4 flux was negatively related to $\delta^{13}\text{C}$ value of emitted CH_4 (Figure 3, $r = -0.875$, $P < 0.01$).

3.4. Contribution of Acetate to Total CH_4 Production (f_{ac}) and Fraction of CH_4 Oxidized (f_{ox})

[21] For comparison with what was obtained by Tyler *et al.* [1997] and Bilek *et al.* [1999], $\delta^{13}\text{CH}_4$ (acetate) was set to be -40‰ and $\alpha_{\text{CO}_2/\text{CH}_4}$ to be 1.079–1.082 [Fey *et al.*, 2004] in equations (5) and (7); furthermore, $\delta^{13}\text{C}$ values of pore water CH_4 (Figure 1b) were used as proxy for total produced CH_4 . It was found that hydrogenotrophic methanogenesis was an important methanogenic pathway over the entire fallow season (Figure 4a), with f_{ac} value being < 0.5 in the early and about 0.6 in the late. After rice transplanting, acetoclastic methanogenesis took the place as the dominant pathway just between July and August, and its f_{ac} value was close to 0.7 but then decreased rapidly toward the end of the rice season. Corresponding to the $\delta^{13}\text{C}$ value of pore water CH_4 , hydrogenotrophic methanogenesis was very important (around 50%) for the most of the season (Figure 4a).

[22] The $\delta^{13}\text{C}$ value of floodwater CH_4 (Figure 1b) in the fallow season was considered to be $\delta^{13}\text{CH}_4$ (oxidized). During the rice season, however, equation (9) and $\epsilon_{\text{transport}} = -12.2\text{‰}$

[Tyler *et al.*, 1997] were used for estimating $\delta^{13}\text{CH}_4$ (oxidized) (why different $\delta^{13}\text{C}$ values were used for $\delta^{13}\text{CH}_4$ (oxidized), please see section 4 below). Moreover, a value of 1.025–1.038 was used for α_{ox} [Coleman *et al.*, 1981; Chanton and Liptay, 2000]. It was calculated in equation (8) that, in terms of percentage, approximately 60%–90% in the fallow season and 10%–80% of CH_4 during the rice season was oxidized before it entered the atmosphere (Figure 4b). Obviously, f_{ox} value was high over the entire fallow season and at the end of rice season but relatively low between July and August. In general, it was higher in the fallow season than in the following rice season (Table 2).

4. Discussion

[23] Numerous measurements show that the carbon isotopic composition of CH_4 from rice fields varies greatly with the methods and types of sampling (Table 1). In principle, CH_4 is more ^{13}C depleted in pore water or gas bubbles from stirred bulk soil than in the aerenchyma of rice plants or in naturally released gas bubbles, because the former is less influenced by CH_4 oxidation [Chanton *et al.*, 1997; Krüger *et al.*, 2002]. Furthermore, aerenchymatic CH_4 is heavier than emitted CH_4 , mainly because of the fact that rice plants preferentially transport $^{12}\text{CH}_4$, making the residual CH_4 ^{13}C enriched in the aerenchyma [Chanton, 2005]. However, these results were mainly about the intermittently flooded rice fields or the rice-growing season (Table 1), and little about rice fields flooded for a whole year. Here we systematically investigated the carbon isotopic composition of CH_4 from a field continuously flooded throughout the fallow and following rice seasons and quantified the pathways of CH_4 production and the fraction of CH_4 oxidized.

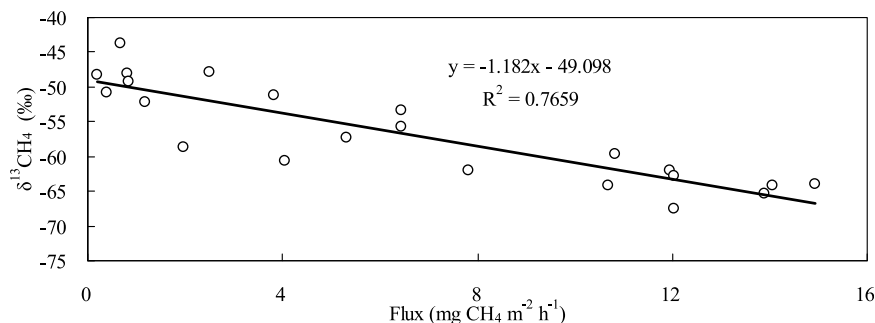


Figure 3. Relationship between flux and corresponding $\delta^{13}\text{CH}_4$ in temporal variation.

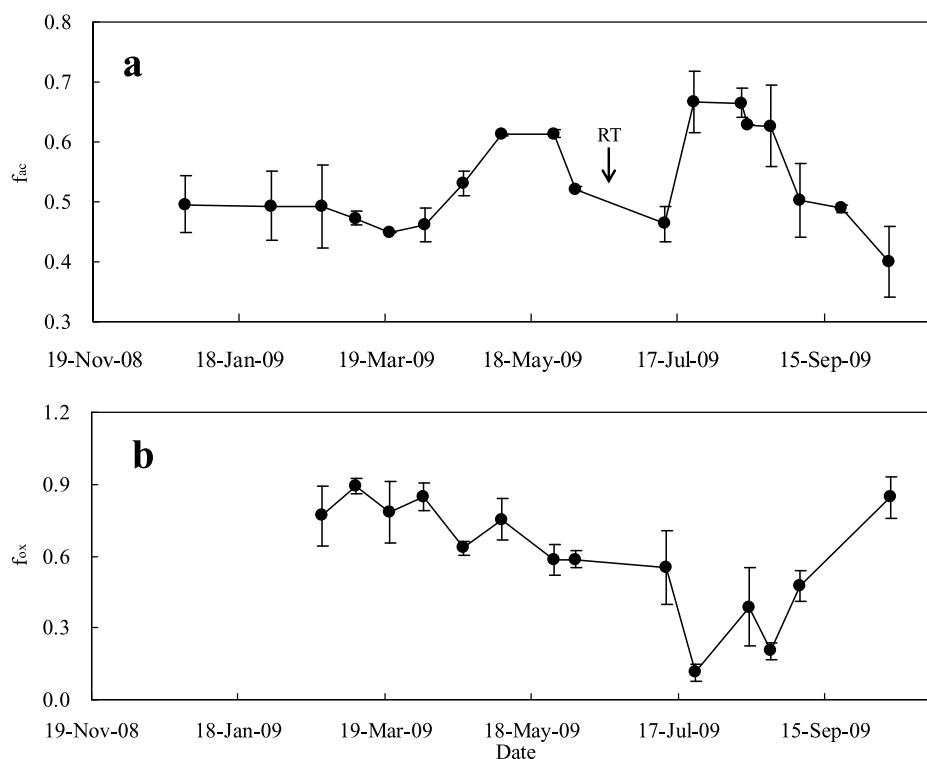


Figure 4. Temporal variations of (a) contribution of acetate to the total CH_4 production (f_{ac}) and (b) the fraction of CH_4 oxidized (f_{ox}) from November 2008 to November 2009 using equations (5) and (8), respectively, and the following parameters: $\delta^{13}\text{C}_4$ (acetate) = -40‰ ; $\alpha_{\text{CO}_2/\text{CH}_4}$ = 1.082 and 1.079 in the fallow season and the following rice season, respectively; α_{ox} = 1.038 and 1.025 in the fallow season and the following rice season, respectively; $\varepsilon_{\text{transport}}$ = -12.2‰ . RT, rice transplanting.

[24] In this study, although the $\delta^{13}\text{C}$ value of pore water CH_4 varied temporally (Figure 1b), it was negative and relatively lighter than most of the earlier measurements (Table 1), which might be attributed to changes of methanogenic pathways. The $\delta^{13}\text{C}$ value of pore water CH_4 varied slightly around -65‰ in the early fallow season, which suggests that acetoclastic methanogenesis during this period was relatively stable, and the f_{ac} value being constantly lower than 0.5 confirmed this. Subsequently, pore water CH_4 was ^{13}C enriched promptly in the late fallow season, indicating an obvious increase in the contribution of acetoclastic methanogenesis (Figure 4a). During the rice season it increased significantly and then remained at a high level between July and August, which is probably ascribed to an increase in CH_4 production via acetate fermentation with the increase in root exudates. The f_{ac} value was found to be

close to 0.7 as well (Figure 4a). Another potential reason would be the rice plants preferential transport ^{12}C of soil pore water into the atmosphere [Popp *et al.*, 1999], because the emitted CH_4 was more ^{13}C depleted than the pore water CH_4 during this period (Figure 1b). Late in the rice season, pore water CH_4 became lighter gradually, indicating that though CH_4 concentration in pore water increased during this period (Figure 1a) probably as a result of plant debris supplying abundant substrates for methanogenesis, it still depended mainly on hydrogenotrophic methanogenesis (Figure 4a).

[25] Carbon isotope fractionation that occurs during the production of CH_4 in paddy soils is very limited to our knowledge [Fey *et al.*, 2004]. The fractionation factor $\alpha_{\text{CO}_2/\text{CH}_4}$, both 1.045 measured by Sugimoto and Wada [1993] in laboratory experiments with Japanese rice field

Table 2. Contribution of Acetate to Total CH_4 Production (f_{ac}) and Fraction of CH_4 Oxidized (f_{ox}) in Comparison With Previous Measurements

Experimental Site	Rice Cultivation	f_{ac}	f_{ox}	Reference
USA	Planted (R) ^a	0.57–0.80 (0.69) ^b	0.19–0.56 (0.39)	Tyler <i>et al.</i> [1997]
	Unplanted (R)	0.51–0.71 (0.62)	0.03–0.21 (0.13)	
USA	Planted (R)	0.62–0.81 (0.73)	0.12–0.71 (0.34)	Bilek <i>et al.</i> [1999]
	Unplanted (R)	0.66	0.24	
China	Planted (R)	0.40–0.67 (0.56)	0.11–0.84 (0.43)	This study
	Unplanted (F)	0.45–0.61 (0.51)	0.59–0.89 (0.73)	

^aR, rice-growing season; F, fallow season.

^bMean value.

soil and 1.06 were considerably used in the latter field conditions [Tyler *et al.*, 1997; Bilek *et al.*, 1999; Krüger *et al.*, 2002]. However, they seemed to be inappropriate to our study. Incubating Italian paddy soils, Fey *et al.* [2004] found that $\alpha_{\text{CO}_2/\text{CH}_4}$ increased with decreasing temperature and was 1.079 and 1.082 at 25 and 15°C, respectively. So, $\alpha_{\text{CO}_2/\text{CH}_4} = 1.082$ in the fallow season and 1.079 during the following rice season is likely to be applicable because of the similar mean temperatures observed in this study (Figure 2). Moreover, Sugimoto and Wada [1993] determined that the $\delta^{13}\text{C}_{\text{CH}_4(\text{acetate})}$ ranged from -43‰ to -30‰ , which was further approved by other studies [Tyler *et al.*, 1997; Bilek *et al.*, 1999; Krüger *et al.*, 2002; Nakagawa *et al.*, 2002]. We used the value of -40‰ because of analogous organic carbon $\delta^{13}\text{C}$ (-26.5‰ in the work of Sugimoto and Wada [1993] and -27.2‰ in the present study) and for better comparing with former reports [Tyler *et al.*, 1997; Bilek *et al.*, 1999]. The results were reasonable and comparable (Figure 4a and Table 2), although more investigations on isotopic fractionation should be made in the further researches.

[26] The process of CH_4 production exhibits seasonal variation in the pathways of methanogenesis, namely, hydrogenotrophic methanogenesis was a very important pathway of CH_4 production for the most of the season, and acetoclastic methanogenesis dominated only between July and August (Figure 4a). It might be related to the differences in soil temperature and rice cultivation. Since the field was unplanted and continuously flooded over the entire fallow season, CH_4 production mainly occurred in the process of decomposition and degradation of endogenous soil organic matters, and a relatively low soil temperature (Figure 2) would be a limited factor for formation of acetate and hence CH_4 production [Cai *et al.*, 2009]. Conrad *et al.* [2009] speculated that the degradation of organic matter which was incompletely mineralized would cause dominance of hydrogenotrophic methanogenesis. When the temperature rose high during the following rice season, especially between July and August (Figure 2), rice planting should become a more important determinant for pathway of methanogenesis [Cai *et al.*, 2009]. On the other hand, investigation of this study found something different from the previous measurements, that is, the contribution of acetoclastic methanogenesis was relatively lower in the rice field that was flooded year-round than in the fields flooded only during the rice-growing season (Table 2). It may be ascribed to the dominance of different methanogenic pathways in regions different in climates and environments, farming and cropping conditions, agricultural managements, rice varieties and soil properties, etc. [Cai *et al.*, 2009].

[27] In the rice fields, CH_4 oxidation occurs mainly in the rhizosphere and at the soil-water interface [Conrad and Rothfuss, 1991; Denier van der Gon and Neue, 1996]. Consequently, the CH_4 dissolved in soil pore water will be partially oxidized when it diffuses into the floodwater. Both Tyler *et al.* [1997] and Bilek *et al.* [1999] found dissolved CH_4 increased in ^{13}C enrichment with the decrease in soil depth. This suggests that for CH_4 produced in the zone, the closer to the soil surface, the more likely to be consumed by methane-oxidizing bacteria. Similarly, the effect of transport on isotopic fractionation would be the other probable reason [Popp *et al.*, 1999]. In our measurements, floodwater CH_4 was more ^{13}C enriched than pore water CH_4 over the season

(Figure 1b), which was in agreement with the measurement of a greenhouse experiment conducted by Cheng *et al.* [2005] during the rice season in Japan.

[28] The CH_4 oxidation is expected to be a key natural process mitigating CH_4 emission from rice fields, and earlier studies through analysis of stable carbon isotopic signatures have shown that 3%–71% of produced CH_4 was oxidized (Table 2). As it is highly influenced by CH_4 oxidation at the soil-water interface, the $\delta^{13}\text{C}$ value of CH_4 in floodwater in the fallow season is regarded as $\delta^{13}\text{C}_{\text{CH}_4(\text{oxidized})}$. During the rice season, however, rice plants become the leading conduit for CH_4 emission, and CH_4 oxidation occurs mainly in the rhizosphere [Denier van der Gon and Neue, 1996; Jia *et al.*, 2001, 2002]. Equation (9) was hence used for estimating $\delta^{13}\text{C}_{\text{CH}_4(\text{oxidized})}$ [Tyler *et al.*, 1997]. Moreover, α_{ox} is significantly affected by temperature and is 1.030–1.033 and 1.034–1.038 at 24 and 12°C, respectively [Chanton and Liptay, 2000; Chanton *et al.*, 2008]. Coleman *et al.* [1981] also found $\alpha_{\text{ox}} = 1.025$ at 26°C, which was later on applied to field data [Tyler *et al.*, 1997; Bilek *et al.*, 1999]. Therefore, it was calculated using $\alpha_{\text{ox}} = 1.038$ in the fallow season and 1.025 during the following rice season that around 60%–90% of CH_4 was oxidized in the fallow season, which might be an important reason for the very low CH_4 flux (Figure 2). Similarly, a relatively lower fraction of CH_4 being oxidized, especially between July and August, was found to be the key factor for the considerable flux during the rice season (Figure 2). Additionally, compared to what was in the former reports, a generally higher fraction of CH_4 oxidized was found (Table 2), which might be interpreted as more CH_4 being emitted by rice plants than by bubble ebullition and molecular diffusion, thus causing less CH_4 oxidation during the rice season than during the fallow season. A previous study showed that CH_4 emission flux was positively related with the efficiency of rice plant-mediated CH_4 transport, but negatively with rhizospheric CH_4 oxidation [Jia *et al.*, 2002].

[29] The $\delta^{13}\text{C}$ value of emitted CH_4 was more positive in the fallow season than most of the available observations conducted during the rice season (Table 1). One possible explanation for the measurements in this study is that a fraction of CH_4 is oxidized. Emitted CH_4 was relatively enriched in ^{13}C , probably because of a high f_{ox} value in the fallow season, whereas a low f_{ox} value in the most part might be a major cause for ^{13}C depletion during the rice season (Figure 1b). With an increase in CH_4 oxidation toward the end of the rice season, $\delta^{13}\text{C}$ values of emitted CH_4 increased again. Krüger *et al.* [2001] found that a seasonal decrease in f_{ox} value resulted in a seasonal decrease in $^{13}\text{CH}_4$. Another probability might lie in the influence of CH_4 transport. CH_4 transport is mainly accomplished through bubble ebullition and molecular diffusion in the fallow season [Tyler *et al.*, 1997], and CH_4 crossing the water–air interface results in low (2–3‰) carbon isotope fractionation [Happell *et al.*, 1994, 1995; Chanton, 2005]. The difference between floodwater and emitted CH_4 in $\delta^{13}\text{C}$ value on 27 May and 5 June was calculated to be 3‰ and 7‰ (Figure 1b), respectively, indicating an analogous fractionation of 3–7‰ in CH_4 transport. On the contrary, CH_4 transport by plants brings about significant carbon isotope fractionation [Chanton, 2005], which would make emitted CH_4 more ^{13}C depleted. Previous studies have showed that CH_4 flux through rice

plants could lead to a fractionation of 11–18‰ by measuring $\delta^{13}\text{C}$ values of emitted and aerenchymatic CH_4 [Chanton et al., 1997; Tyler et al., 1997; Bilek et al., 1999; Krüger et al., 2002; Krüger and Frenzel, 2003]. Another experiment by the authors found a similar result (G. Zhang et al., unpublished data). Furthermore, flux was negatively related to $\delta^{13}\text{C}$ value of emitted CH_4 in temporal variation (Figure 3), suggesting that plant transport is probably a more important pathway and less CH_4 oxidation occurs during the rice season than in the fallow season.

[30] After the final drainage, a sharp fluctuation in $\delta^{13}\text{C}$ value of emitted CH_4 should be paid considerable attention to (Figure 1b). It drastically increased 1 day after the drainage, which is mainly ascribed to the influence of CH_4 oxidation. It is expected that draining of the rice field results in ^{13}C enrichment [Tyler et al., 1994; Bergamaschi, 1997; Marik et al., 2002; Han et al., 2005; Rao et al., 2008] because of increased activity of methanotrophs both in and around the rhizosphere. Two days later, a significant decrease in $\delta^{13}\text{C}$ value of emitted CH_4 was observed. A possible explanation for this might be that a substantial amount of soil-entrapped CH_4 was released instantaneously at that time (Figure 2) and too quickly to be influenced by oxidation. The emitted CH_4 was ^{13}C enriched again as the days after drainage went on, which is most likely attributable to CH_4 oxidation as well.

5. Conclusions

[31] By measuring $\delta^{13}\text{C}$ values of different CH_4 pools in a continuously flooded rice field throughout the fallow and following rice seasons, it was found that pore water CH_4 was more ^{13}C depleted over the season except between July and August; The $\delta^{13}\text{C}$ value of CH_4 was more positive in floodwater (from -50‰ to -34‰) than in pore water (from -68‰ to -54‰) over the season, which indicates that floodwater CH_4 is highly oxidized; $\delta^{13}\text{C}$ value of emitted CH_4 was negatively related to flux in temporal variation. In addition, it was heavier in the fallow season (between -56‰ and -44‰) than in the rice season (between -68‰ and -48‰). This demonstrates that CH_4 transport through rice plants is more important than through molecular diffusion, which brings about less methanotrophy during the rice season than during the fallow season.

[32] Further calculation of methanogenic pathway and fraction of CH_4 oxidized by using isotopic data shows that hydrogenotrophic methanogenesis was very important for the most of the season, and acetoclastic methanogenesis became dominant only between July and August during the period of vigorous growth of rice plants; Approximately 60%–90% of produced CH_4 was oxidized in the fallow season, while only 10%–80% during the following rice season. In short, a low contribution of acetoclastic methanogenesis and a high fraction of CH_4 oxidized in the field were found, especially in the fallow season.

[33] **Acknowledgments.** The authors are grateful to Yong Han and Yacheng Cao, etc., at the Institute of Soil Science, Chinese Academy of Sciences, Nanjing, China, for their help in isotopic data analysis. The authors are also grateful to our local collaborators for field assistance and to our anonymous reviewers for their helpful comments on the article. Funding for this research was provided by the National Natural Sciences Foundation of China (grants 40921061 and 40971154) and the Ministry of Science and Technology of China (S2010GR0080).

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